



ARTICLE



<https://doi.org/10.1057/s41599-026-06688-0>

OPEN

Multidimensional ergonomic latent modeling of prefabricated adaptive decoration by robotic 3D printing

Gangwei Cai^{1,2,3✉}, Binyan Xu⁴, Zehuan Hu⁵, Zhenwei Guo^{6,7,8✉}, Sheng He⁹, Feidong Lu^{10✉}, Yiping Xia², Ziming Ren^{11,12✉}, Bart Julien Dewancker⁴ & Weijun Gao^{4,13,14}

Chain hotels constitute 40.09% of all hotels, with indoor room decorations typically undergoing replacement cycles every 2 to 5 years. Robotic 3D printing-enabled prefabrication offers transformative potential for sustainable construction, combining precision engineering with environmental consciousness. This study explores its application in the hospitality sector, addressing both sustainability and ergonomic data. Using AI-driven multidimensional ergonomic latent modeling, key environmental factors such as sound insulation, lighting adaptability, and air quality are evaluated for their impact on service flexibility, safety, and user experience. By aligning technological innovation, this research provides a latent model for advancing prefabricated building adaptive decoration and low-carbon hotel design. The study underscores the importance of interdisciplinary efforts to achieve global sustainability goals and redefines prefabrication technologies, especially robotic 3D printing prefabricated decoration, as catalysts for operational excellence and transformative industry standards in sustainable hospitality. Moreover, this work extends the exploration of robotic 3D printing prefabricated solutions to the context of extreme environments and even extraterrestrial buildings, such as building habitats on Mars or the Moon, contributing valuable human factors engineering insights for future adaptive built environment research in these challenging settings.

¹Department of Architecture, Graduate School of Engineering, The University of Tokyo, Tokyo, Japan. ²Institute of Landscape Architecture, Zhejiang University, Hangzhou, China. ³Hangzhou International Urbanology Research Center & Zhejiang Urban Governance Studies Center, Hangzhou, China. ⁴Faculty of Environmental Engineering, University of Kitakyushu, Fukuoka, Japan. ⁵The Center for Energy Systems Design (CESD), Kyushu University, Fukuoka, Japan. ⁶China Academy of Building Research, Beijing, China. ⁷Chinese Society for Urban Studies, Beijing, China. ⁸School of Architecture, Tianjin University, Tianjin, China. ⁹School of Art, Southeast University, Nanjing, China. ¹⁰Tongji Architectural Design (Group) Co., Ltd., Shanghai, China. ¹¹School of Civil Engineering and Architecture, Zhejiang Sci-Tech University, Hangzhou, China. ¹²Zhejiang Key Laboratory of Green, Digital and Intelligent (GDI) Renovation for Urban Infrastructures, Hangzhou, China. ¹³Institut Teknologi Sepuluh Nopember, Kampus ITS Sukolilo Surabaya, Surabaya, Indonesia. ¹⁴The Engineering Academy of Japan, Tokyo, Japan. ✉email: cai.gangwei@maelab.arch.t.u-tokyo.ac.jp; Marsswok@tju.edu.cn; feidonglu@tongji.edu.cn; zimingren@zju.edu.cn

Introduction

The global shift toward sustainable development and rising consumer environmental awareness has increased interest in green building practices. Chain hotels accounted for 40.09% of the total number of hotels in China (2025 China Hotel Industry Development Report: Seizing Opportunities in Cyclical Patterns n.d.), and their indoor room decorations were replaced every 2–5 years (Cai, Liu, et al., 2025; Mostafavi et al., 2021). The resource-intensive hospitality industry is adopting eco-friendly approaches, but research on integrating low-carbon technologies with building design, particularly from a tourism perspective, is limited (Li, 2023). This study focuses on budget chain hotels, an often-overlooked sector, to enhance tourist satisfaction through prefabricated room designs that align with low-carbon principles and sustainable building environments. It supports sustainable development goals (SDGs) 12.5 (waste reduction) and 8.9 (sustainable tourism), addressing environmental and economic sustainability (Cai, Wang, et al., 2023). Prefabricated hotel room designs can reduce construction waste, improve energy efficiency, and enhance guest experiences through the integration of ergonomics and human factors engineering (Kaifosh et al., 2025), ensuring that design solutions are not only sustainable but also optimized for user comfort, safety, and operational efficiency (Shi et al., 2025), addressing environmental concerns and meeting rising demand for sustainable tourism (Bian et al., 2024). Among these innovations, green prefabrication technologies, particularly those leveraging robotic 3-dimensional (3D) printing, have emerged as a transformative solution to reimagine conventional construction paradigms (Dong et al., 2024).

Robotic 3D printing-enabled prefabrication represents an AI-driven cutting-edge integration of precision engineering and environmental sustainability (Cai, Sun, et al., 2025; Raphael et al., 2023). This technology enables efficient modular component production, reducing construction waste, optimizing material use, and supporting adaptive designs for diverse environmental and functional requirements (Gündüz & Özkaz, 2024; Sigahi et al., 2024). These advancements hold particular significance for the hospitality sector, where sustainability is increasingly prioritized by consumers and policymakers (Badraoui & Smith, 2024). Hotels, as substantial energy users and carbon emitters, stand to benefit significantly from incorporating green prefabrication strategies into their design and operations (Xu et al., 2022). Despite these advancements, challenges persist in scaling robotic 3D printing technologies within the hospitality industry (Guanán-Rivera et al., 2022). Existing research predominantly emphasizes technical improvements, such as material performance and energy efficiency, while neglecting the relationship between these innovations and customer satisfaction (Azzaoui & Dyba, 2022; Çallı et al., 2022). Moreover, economic feasibility and regulatory barriers remain underexplored, hindering widespread adoption (Arenhart et al., 2024). Addressing these gaps requires a comprehensive framework that aligns technological innovation with environmental goals and market demands (Živković et al., 2023).

This study develops a holistic model to analyze the relationship between green prefabrication technologies and tourist satisfaction. Using advanced statistical methods, including confirmatory factor analysis (CFA) and structural equation modeling (SEM), the research evaluates the impact of environmental quality factors—such as sound insulation, lighting adaptability, and air quality—on service flexibility, safety, and guest satisfaction (Kyriazos & Poga-Kyriazou, 2023; Xia & Zhang, 2022). By bridging the gap between innovation and user experience, the study provides actionable insights for industry stakeholders and policymakers (Puzatova et al., 2022). The findings contribute to sustainable construction discourse by presenting a framework for integrating

robotic 3D printing technologies into low-carbon hotel design (Lohakare et al., 2024). Beyond mitigating environmental impact, the framework demonstrates the potential of these technologies to redefine industry standards and drive transformative change. This research aims to catalyze a paradigm shift in sustainable construction, highlighting prefabrication technologies—particularly robotic 3D printing and prefabricated decoration—as tools for reducing environmental footprints while enhancing user satisfaction and operational performance. Notably, the study also extends the application of robotic 3D printing prefabrication to the exploration of extreme environments and even extraterrestrial architecture, such as Martian habitats or lunar bases, by contributing valuable insights from the perspective of human perception. By addressing critical gaps in existing literature and providing a forward-looking perspective, this study lays the groundwork for future advancements in green prefabrication and sustainable hospitality, as well as innovative construction strategies for challenging and uncharted settings.

Literature review

Recent studies have highlighted that integrating ergonomics and human factors engineering into prefabricated building design can enhance spatial usability, occupant comfort, and operational efficiency, while complementing low-carbon and sustainable construction objectives (Cheng et al., 2022; Sigahi et al., 2024). The integration of robotic 3D printing into prefabrication processes has emerged as a pivotal innovation, allowing for precise customization, material efficiency, and reduced waste (Bao et al., 2023). Some previous studies demonstrated how robotic 3D printing techniques significantly improve the environmental performance of building components, particularly in insulation, ventilation, and natural lighting systems (Xue et al., 2024). These advancements align with global SDGs by promoting energy efficiency and minimizing resource depletion (Abdollahi et al., 2023; Tran et al., 2023). Green prefabrication technologies also address critical environmental challenges associated with traditional construction methods (Mo et al., 2023; Parisi et al., 2023; Tran et al., 2025). Prefabricated modular designs enabled by robotic 3D printing have been shown to reduce construction time and energy consumption while enhancing structural quality (Cai, Liu, et al., 2025; Pollák et al., 2024). For example, studies on thermal comfort and air quality (TCAQ) suggest that robotic 3D printing modules offer superior performance in regulating indoor environments, thereby contributing to occupant health and well-being (Abdullah et al., 2024). These findings underscore the role of advanced prefabrication techniques in creating energy-efficient and sustainable building systems (Jing et al., 2022).

The application of green prefabrication technologies within the hospitality industry has gained traction as hotels strive to align with sustainability goals (Cai, Guo, et al., 2025; Geng et al., 2023). Research has shown that adopting green building standards not only reduces Leadership in Energy and Environmental Design (LEED) environmental impact but also enhances guest satisfaction by delivering superior indoor environments (Cai et al., 2021; Kim et al., 2024). For instance, studies on sound insulation, lighting adaptability, and air quality improvements have demonstrated that these features play a critical role in shaping guest experiences (Xu et al., 2022). The incorporation of these standards into hotel design has been linked to increased market competitiveness and long-term profitability (Ordoñez-Ávila et al., 2023). Furthermore, green hospitality practices are increasingly guided by frameworks such as LEED and Building Research Establishment Environmental Assessment Method (BREEAM) (Jiayue et al., 2024). These frameworks emphasize energy

efficiency, water conservation, and material sustainability (Rafsanjani, 2022). However, the literature highlights a gap in adapting these standards to the unique requirements of prefabricated hotel designs (Ho, 2022). Addressing this gap is crucial for advancing the integration of green prefabrication technologies into the hospitality sector (Ostapska et al., 2024).

Guest satisfaction has emerged as a key metric for evaluating the success of green prefabrication technologies in the hospitality industry (Çallı et al., 2022). Studies indicate that features such as flexible space layouts, health-promoting materials, and eco-friendly amenities significantly influence guest perceptions (Badraoui & Smith, 2024; Cai, Lou, et al., 2025; Shan et al., 2023). For example, robotic 3D printing-enabled modular designs have been shown to facilitate personalized services and adaptive environments, which are highly valued by guests (Bedarf et al., 2023). These innovations not only enhance guest satisfaction but also contribute to the industry's sustainability goals (Cai, Xu, et al., 2023; Prianto et al., 2022). Despite these advancements, unresolved issues remain in optimizing green prefabrication technologies for maximum impact. Research has yet to fully explore the economic feasibility of implementing these technologies on a large scale (Cai, Zou, et al., 2023; Kanthimathi et al., 2024). Additionally, there is a need for more comprehensive studies on the long-term performance of green prefabrication systems in varying environmental conditions (Liu et al., 2023). Addressing these challenges will require interdisciplinary collaboration and the development of new frameworks that integrate environmental, economic, and social considerations.

Although green prefabrication technologies have gained attention in recent years, a critical gap persists in quantitatively assessing how robotic 3D printing for prefabricated decoration influences guest satisfaction in chain hotel environments. Existing literature has predominantly focused on environmental or operational benefits, often overlooking the user-centered impact of such technologies, especially in relation to indoor environmental quality factors such as acoustic comfort, air quality, and lighting adaptability. This study addresses this gap by integrating confirmatory factor analysis and structural equation modeling (CFA-SEM) to evaluate how these environmental elements, facilitated by robotic prefabrication, affect tourist satisfaction.

Methods

This study adopts a structured methodology to investigate the relationship between green prefabricated hotel room features and tourist satisfaction (Fig. 1). The research framework is grounded in both international and national low-carbon and prefabrication standards. These standards provide the foundational criteria for evaluating sustainability and interior environmental performance

in hotel buildings. Building upon these standards, a comprehensive literature review was conducted to identify key dimensions influencing guest satisfaction in green and healthy prefabricated hotel rooms. Relevant empirical studies were analyzed to extract established indicators across domains such as AI-driven renovation quality, aesthetic design, indoor environmental quality, and perceived sustainability. By integrating insights from both standards and prior research, the study formulated a set of measurement items corresponding to latent variables, including AI-enhanced renovation, low-carbon design features, aesthetic quality, comfort and functionality, perceived sustainability, tourist satisfaction, and loyalty. These items were operationalized into a structured questionnaire using a 7-point Likert scale. Subsequently, a series of hypotheses (H1–H5) was developed to describe the conceptual relationships among the constructs, forming the basis of the hypothetical model. To validate this model, data collected from hotel guests were analyzed using Confirmatory Factor Analysis (CFA) to assess construct validity and measurement reliability. Then, Structural Equation Modeling (SEM) was employed to test the proposed hypotheses and examine the direct and mediating effects of AI-enhanced prefabrication on tourist satisfaction and loyalty. This quantitative approach provides a robust framework to explore the interplay between prefabricated construction strategies and sustainable tourism outcomes.

Standards. Global low-carbon hotel standards (Table 1) include LEED, BREEAM, Green Key, Earth Check, Global Sustainable Tourism Council (GSTC), Energy Star, and TripAdvisor Green Leaders. LEED, developed by USGBC, focuses on energy savings and carbon reduction, offering Certified, Silver, Gold, and Platinum levels (Gupta & Dantsiou, 2013; Peng et al., 2023; Wei et al., 2024; Wong et al., 2015; Ye et al., 2023). BREEAM, from the UK, evaluates sustainability across energy, water, materials, and pollution. Green Key emphasizes sustainable management, recognized by GSTC. Earth Check benchmarks environmental impacts, popular in the Asia-Pacific. GSTC sets global sustainability criteria. Energy Star rates energy efficiency, while TripAdvisor Green Leaders highlights eco-friendly practices. Standards like JGJ/T491-2021 and T/JCJJ002-2023 enhance prefabricated interior quality and efficiency, promoting low-carbon practices in hotels to achieve sustainability goals (Chan, 2012).

References. Customer satisfaction in green hotels is increasingly linked to environmental responsibility. Post-pandemic, travelers are more conscious of the sustainability practices of the places they stay. Research shows that green hotels, especially those employing prefabricated designs, have a competitive advantage

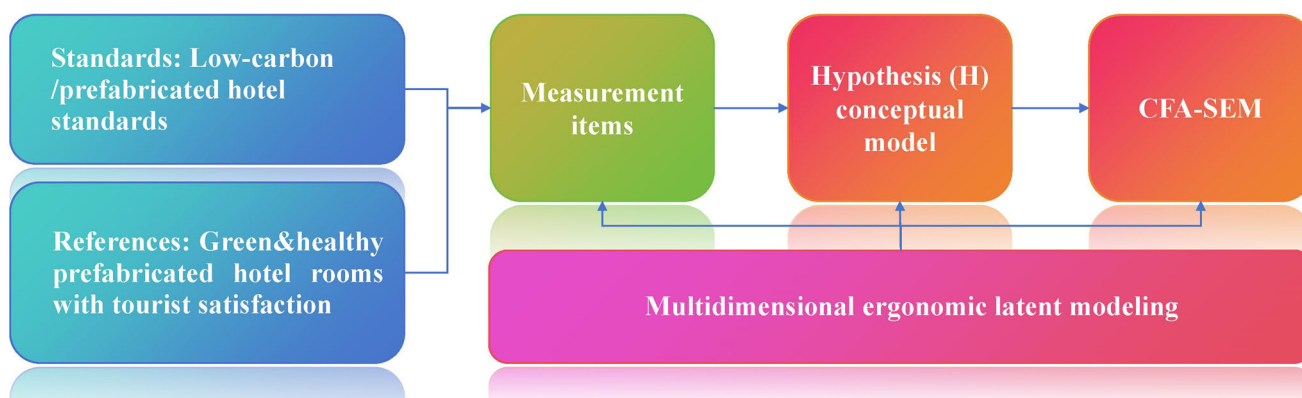


Fig. 1 Methodology process.

Table 1 Standards.			
Standard	Origin/Organization	Focus/Criteria	Additional Notes
LEED	USGBC (USA)	Energy savings, carbon reduction	Levels: Certified, Silver, Gold, Platinum
BREEAM	UK	Energy, water, materials, pollution	Comprehensive sustainability evaluation
Green Key	Recognized by GSTC	Sustainable management	International recognition
Earth Check	Asia-Pacific	Environmental benchmarking	Popular in Asia-Pacific region
GSTC	Global	Sustainability criteria	Sets global baseline standards
Energy Star	USA	Energy efficiency	Applies to buildings and appliances
TripAdvisor Green Leaders	TripAdvisor	Eco-friendly practices	Highlights hotels with green practices
JGJ/T491-2021	China	Prefabricated interior quality	Promotes low-carbon practices
T/JCJ002-2023	China	Efficiency in prefabrication	Supports hotel sustainability goals

due to their lower environmental impact and enhanced health and safety measures. The recovery of the hospitality industry, therefore, hinges on adopting green building practices that align with customer expectations for sustainability and well-being. The design of hotel rooms plays a pivotal role in achieving both sustainability goals and guest satisfaction. Prefabricated hotel rooms, in particular, can be designed to meet the highest standards of environmental and health performance, contributing significantly to tourist satisfaction (Mohandes et al., 2022).

Acoustics: Prefabricated hotels use optimized materials for acoustic insulation, enhancing guest comfort and reducing noise pollution while improving energy efficiency (Dong et al., 2024; Qiao et al., 2023). **Lighting & illumination & View:** Prefabricated hotels integrate natural and energy-efficient lighting, improving ambiance and reducing energy use, with smart controls and well-designed windows enhancing guest satisfaction and well-being (Raphael et al., 2023). **Thermal comfort & Air quality:** Advanced prefabrication enhances thermal comfort and air quality through improved insulation and smart HVAC, boosting guest well-being and reducing energy use (Walczyk & Ożadowicz, 2024; Živković et al., 2023).

Flexible Service ability: Prefabricated hotel rooms offer flexibility and customization, adapting quickly to guest needs with modular furniture and reconfigurable layouts, enhancing service while reducing waste and resource consumption (Ostapska et al., 2024; Zhang et al., 2022). **Health, safety, and security:** Prefabrication improves hotel safety and hygiene with non-toxic materials, advanced construction, ventilation, and antimicrobial surfaces, addressing health concerns heightened by the pandemic (Hong et al., 2020). **Green service:** Prefabricated hotels using renewable energy, water-saving tech, and green services appeal to eco-conscious travelers, aligning with environmental values and enhancing guest experience (Lertputtarak et al., 2023; Puzatova et al., 2022).

The development of green standards for low-carbon and prefabricated hotels has been a growing focus. Research has emphasized the need for integrating sustainability into every aspect of building design and construction. The insights gained from these surveys have informed the development of more robust standards and guidelines for the industry. Measurement items in Table 2 of prefabricated room decoration for the questionnaire survey on customer satisfaction were identified.

Hypothesis (H) conceptual model. Based on previous papers about green and tourist satisfaction, Hypothesis (H) was shown in the proposed conceptual model: Sound environment (SE), Lighting & illumination& View(LIV), Thermal comfort& Air quality (TCAQ), Flexible serviceability (FS), Health, safety, and security (HSS), Green service (GS), and Tourist satisfaction (TS).

Sample testing model for CFA-SEM. CFA-SEM was used to test the hypothetical model : (1) a measurement model to evaluate the

component scores was implemented; (2) CFA was used to check the validity; (3) SEM was used to evaluate the model.

Equations (1)–(3) show the measurement model:

$$X_i = \lambda_i F_1 + e_i; i = \overline{1, 3} \tag{1}$$

$$X_j = \lambda_j F_2 + e_j; j = \overline{4, 6} \tag{2}$$

$$X_k = \lambda_k F_3 + e_k; k = \overline{7, 9} \tag{3}$$

where the variables X_1, X_2, X_3 , and the coefficient F_1 linearly depend on the coefficients $\lambda_1, \lambda_2, \lambda_3$, and e_1, e_2, e_3 is its measurement error. Similarly, regression (2), (3) is the linear relationship of X_4, X_5, X_6 , and F_2 , and X_7, X_8, X_9 , and F_3 .

The structural model is given by Eq. (4):

$$F_3 = \beta_1 F_1 + \beta_2 F_2 + e_{10} \tag{4}$$

where β_1, β_2 are the regression coefficients between factors F_1, F_2 , and F_3 (with potential errors e_{10}).

Bootstrapping (maximum likelihood estimation) is used to test the assumption. Maximum likelihood estimation (MLE) aims to produce a predicted covariance matrix $\hat{\Sigma}$ (as close as possible to the sample covariance matrix $\bar{\Sigma}$). The difference between $\hat{\Sigma}$ & $\bar{\Sigma}$ is the minimization of the fitting function. Equation (5) is solved by an iterative procedure with a selected starting value:

$$f_{ML} = \ln |\hat{\Sigma}| - \ln |\bar{\Sigma}| + \text{tr} \left[\bar{\Sigma} \hat{\Sigma}^{-1} \right] - p \tag{5}$$

Where: p is the number of variables; determinants and traces summarize important information about the matrix $\hat{\Sigma}$ & $\bar{\Sigma}$.

In addition, f_{ML} is used in the chi-squared (χ^2 , goodness-of-fit) test, which measures how well the model fits the sample (under the assumption of normality, the model is used to fit the exponent) (6):

$$X^2 = f_{ML} \times (n - 1) \tag{6}$$

Where n is the number of samples.

Results

The photograph of the word ‘H’ shows the Hypothesis.

H1: The presence of a sound environment (SE) positively influences flexible serviceability (FS).

H2: The presence of a sound environment (SE) positively influences health, safety, and security (HSS).

H3: The presence of lighting & illumination& View(LIV) positively influences flexible serviceability (FS).

H4: The presence of lighting & illumination& view(LIV) positively influences green service (GS).

H5: The presence of thermal comfort& air quality (TCAQ) positively influences flexible serviceability (FS).

H6: The presence of thermal comfort& air quality (TCAQ) in positively influences health, safety, and security (HSS).

Table 2 Measurement items.

Items	Number
Robotic 3D printing driven Green indoor environment	Q1
Robotic 3D printing driven Sound environment (SE) (Dong et al., 2024; Qiao et al., 2023)	Q1-1
Walls, doors, and windows fabricated with robotic 3D printing techniques significantly improve room sound insulation.	1
The sound absorption performance of wall decoration materials produced using robotic 3D printing modules is highly satisfactory.	2
Robotic 3D printing driven Lighting & illumination & View (LIV) (Cai et al., 2020; Raphael et al., 2023)	Q1-2
Robotic 3D printing technology effectively adapts room-size windows to provide an ideal external view.	3
Indoor brightness levels are easily adjustable due to the use of robotic 3D printing module materials.	4
Anti-glare and shading measures facilitated by prefabricated robotic 3D printing designs are highly effective.	5
Robotic 3D printing modules enhance daylight utilization in an efficient manner.	6
Robotic 3D printing driven Thermal comfort& Air quality (TCAQ) (Walczyk & Ożadowicz, 2024; Živković et al., 2023)	Q1-3
Robotic 3D printing-based modular designs provide effective humidity control.	7
Robotic 3D printing module materials maintain indoor air quality at a high standard.	8
Adaptive prefabricated designs enabled by robotic 3D printing improve natural ventilation.	9
Green modules produced by robotic 3D printing allow air conditioning systems to deliver fresh air and enhance air quality.	10
Room temperature is well-regulated through robotic 3D printing modular designs.	11
The smoking room floors utilize robotic 3D printing technology to incorporate anti-smoking module materials	12
Robotic 3D printing -driven Green indoor service quality	Q2
Robotic 3D printing driven Flexible service ability (FS) (Ostapska et al., 2024; Zhang et al., 2022)	Q2-1
Personalized, adaptive bedding options produced through robotic 3D printing designs meet guest needs.	13
Health-promoting plants placed using robotic 3D printing green modules are highly beneficial.	14
Smart devices integrated into robotic 3D printing-based prefabricated designs are highly functional and user-friendly.	15
Maintenance is efficiently managed through robotic 3D printing green indoor materials.	16
Flexible space layouts are achieved through robotic 3D printing adaptive prefabricated designs.	17
Robotic 3D printing driven Health, safety, and security (HSS) (Cai, Zhang, et al., 2023; Mo et al., 2022; Ostapska et al., 2024)	Q2-2
Electrical switches installed through robotic 3D printing adaptive designs are fully functional and clearly labeled.	18
Robotic 3D printing green module materials enhance natural disaster risk management.	19
Robotic 3D printing green module materials demonstrate high reliability.	20
Chemical and biological pollutants are effectively addressed using robotic 3D printing green module materials.	21
Smoke alarms and automatic sprinklers installed with robotic 3D printing green module materials provide adequate safety measures.	22
Robotic 3D printing prefabricated designs effectively incorporate crime prevention measures.	23
Robotic 3D printing driven Green service (GS) (Lertputtarak et al., 2023; Puzatova et al., 2022)	Q2-3
Clean drinking water is supplied through robotic 3D printing green module materials.	24
Consumables produced using robotic 3D printing green module materials are recyclable and satisfactory.	25
Cotton fabrics made from robotic 3D printing green module materials are neat and have excellent moisture absorption.	26
Replacement reminder cards for cotton fabrics, provided through robotic 3D printing green module materials, are helpful.	27
Energy-saving reminder cards on robotic 3D printing green module materials contribute to environmental conservation.	28
Tourist satisfaction (Badraoui & Smith, 2024; Çallı et al., 2022; Yang, 2024)	Q3
I think robotic 3D printing prefabricated room decoration is an important part of sustainable tourism	29
I am satisfied with the robotic 3D printing prefabricated room decoration.	30

H7: The presence of a sound environment (SE) positively influences tourist satisfaction (TS).

H8: The presence of lighting & illumination& view(LIV) positively influences tourist satisfaction (TS).

H9: The presence of flexible serviceability (FS) positively influences green service (GS).

H10: The presence of health, safety, and security (HSS) positively influences tourist satisfaction (TS).

H11: The presence of green service (GS) positively influences tourist satisfaction (TS).

The measurement model was evaluated using CFA supported by integrated visual analyses of structural relations, construct topology, and correlational patterns (Fig. 2). The data fit of the measurement model is acceptable. The visual analyses collectively reveal the structural properties of the measurement and structural models. The β -r dumbbell comparison shows that several paths demonstrate high convergence between structural effects and bivariate correlations, notably H7 (SE $\rightarrow$ TS: $\beta = 0.70$; $r = 0.90$) and H5 (TCAQ $\rightarrow$ FS: $\beta = 0.90$; $r = 0.82$). In contrast, other paths exhibit strong correlations but weaker structural effects, such as H11 (GS $\rightarrow$ TS: $\beta = 0.14$; $r = 0.92$) and H6 (TCAQ $\rightarrow$ HSS: $\beta = 0.06$; $r = 0.60$), indicating mediation and shared variance among constructs. The MDS/PCA construct map demonstrates clear clustering: F1–F3 (SE, LIV, TCAQ) form a compact

environmental quality cluster, driven by high intercorrelations (0.62–0.94), whereas F5–F6 (HSS, GS) form a service–safety cluster ($r = 0.26$ –0.42). F7 (TS) emerges near SE and GS, reflecting their strong associations. The lower-triangle correlation matrix highlights the strongest relationships: F2–F3 ($r = 0.94$), GS–TS ($r = 0.92$), SE–TS ($r = 0.75$). Moderate associations link FS, HSS, and GS (0.26–0.41), while lower correlations (0.24–0.35) indicate distinct variance within service–safety constructs. The clustered heatmap confirms two dominant latent dimensions: environmental comfort (F1–F3) and green service–safety (F4–F6). F7 (TS) operates as an integrative endpoint connecting both clusters. Collectively, these visual diagnostics show that AI-enhanced prefabricated environmental features and green service attributes jointly shape tourist satisfaction, with environmental constructs exhibiting the strongest structural coherence.

Figure 3 display the results of SEM. The proposed structural model is suitable for the data. The SEM results show that all 11 hypotheses are statistically supported at the 0.05 level (Jeong & Hyun, 2019): H1: The presence of sound environment (SE) in positively influences flexible serviceability (FS); H2: The presence of sound environment (SE) in positively influences health, safety, and security (HSS); H3: The presence of lighting & illumination& View(LIV) in positively influences flexible serviceability (FS); H4: The presence of lighting & illumination& view(LIV) in positively

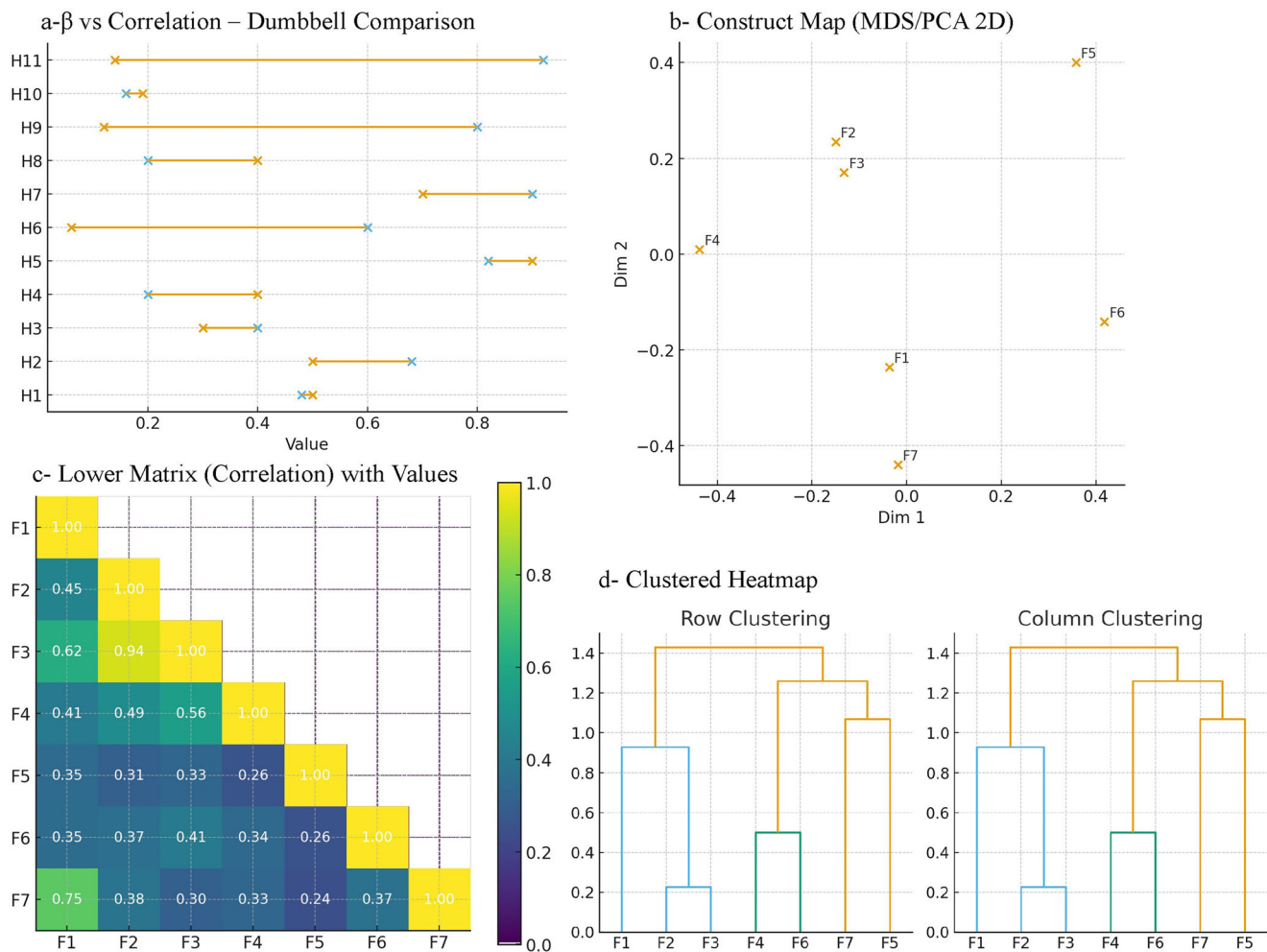


Fig. 2 Integrated visualization of structural relations, construct topology, and correlational architecture. **a** β - r dumbbell comparison illustrating divergences between structural effects and bivariate associations. **b** Two-dimensional MDS/PCA construct map showing latent-space proximities among F1-F7. **c** Lower-triangle correlation matrix with coefficients highlighting the magnitude and symmetry of inter-construct relationships. **d** A clustered heatmap with hierarchical row and column dendrograms revealing data-driven construct groupings.

influences green service (GS); H5: The presence of thermal comfort& air quality (TCAQ) in positively influences flexible serviceability (FS); H6: The presence of thermal comfort& air quality (TCAQ) in positively influences health, safety, and security (HSS); H7: The presence of sound environment (SE) in positively influences tourist satisfaction (TS); H8: The presence of lighting & illumination& view(LIV) in positively influences tourist satisfaction (TS); H9: The presence of flexible serviceability (FS) in positively influences green service (GS); H10: The presence of health, safety, and security (HSS) in positively influences tourist satisfaction (TS); H11: The presence of green service (GS) in positively influences tourist satisfaction (TS).

A multidimensional evaluation of structural effects, correlational patterns, and construct contributions was used (Fig. 4). The path-importance ranking shows that H5 (TCAQ $\rightarrow$ FS) exhibits the strongest structural effect ($\beta = 0.90$), followed by H7 (SE $\rightarrow$ TS; $\beta = 0.70$). Medium effects include H1 (SE $\rightarrow$ FS) and H2 (SE $\rightarrow$ HSS) (both $\beta = 0.50$), H4 (LIV $\rightarrow$ GS) and H8 (LIV $\rightarrow$ TS) (both $\beta = 0.40$), whereas H3 (LIV $\rightarrow$ FS; $\beta = 0.30$), H10 (HSS $\rightarrow$ TS; $\beta = 0.19$), H11 (GS $\rightarrow$ TS; $\beta = 0.14$), H9 (FS $\rightarrow$ GS; $\beta = 0.12$), and H6 (TCAQ $\rightarrow$ HSS; $\beta = 0.06$) reflect weaker influences. The β - r dumbbell comparison reveals strong agreement for H7 ($\beta = 0.70$; $r = 0.90$) and H5 ($\beta = 0.90$; $r = 0.82$), but substantial discrepancies for H11 ($\beta = 0.14$; $r = 0.92$) and H6 ($\beta = 0.06$;

$r = 0.60$), indicating suppression and mediation effects within the SEM. The ridge plot demonstrates a skewed distribution of structural coefficients, with most β values clustering between 0.10–0.50, a dominant peak at $\beta = 0.90$, and only one minimal effect ($\beta = 0.06$), confirming a model driven by one major path (TCAQ $\rightarrow$ FS) and several moderate contributors. Direct determinants of tourist satisfaction (TS) include SE ($\beta = 0.70$; 49%), LIV ($\beta = 0.40$; 28%), HSS ($\beta = 0.19$; 13%), and GS ($\beta = 0.14$; 10%), showing that environmental comfort—especially acoustic quality—is the primary influence. The lollipop chart visually reinforces this ranking, highlighting the steep decline from SE to the remaining predictors and demonstrating SE's dominant role relative to visual comfort, safety, and green service attributes.

Discussion

This study delves into the dynamic relationship between green prefabrication technologies, tourist satisfaction, and sustainable practices in the hospitality industry. By integrating advanced robotic 3D printing techniques, the research contributes to a deeper understanding of how green and low-carbon standards in hotel design can enhance environmental performance and service quality. The findings provide valuable insights into the theoretical and practical dimensions of

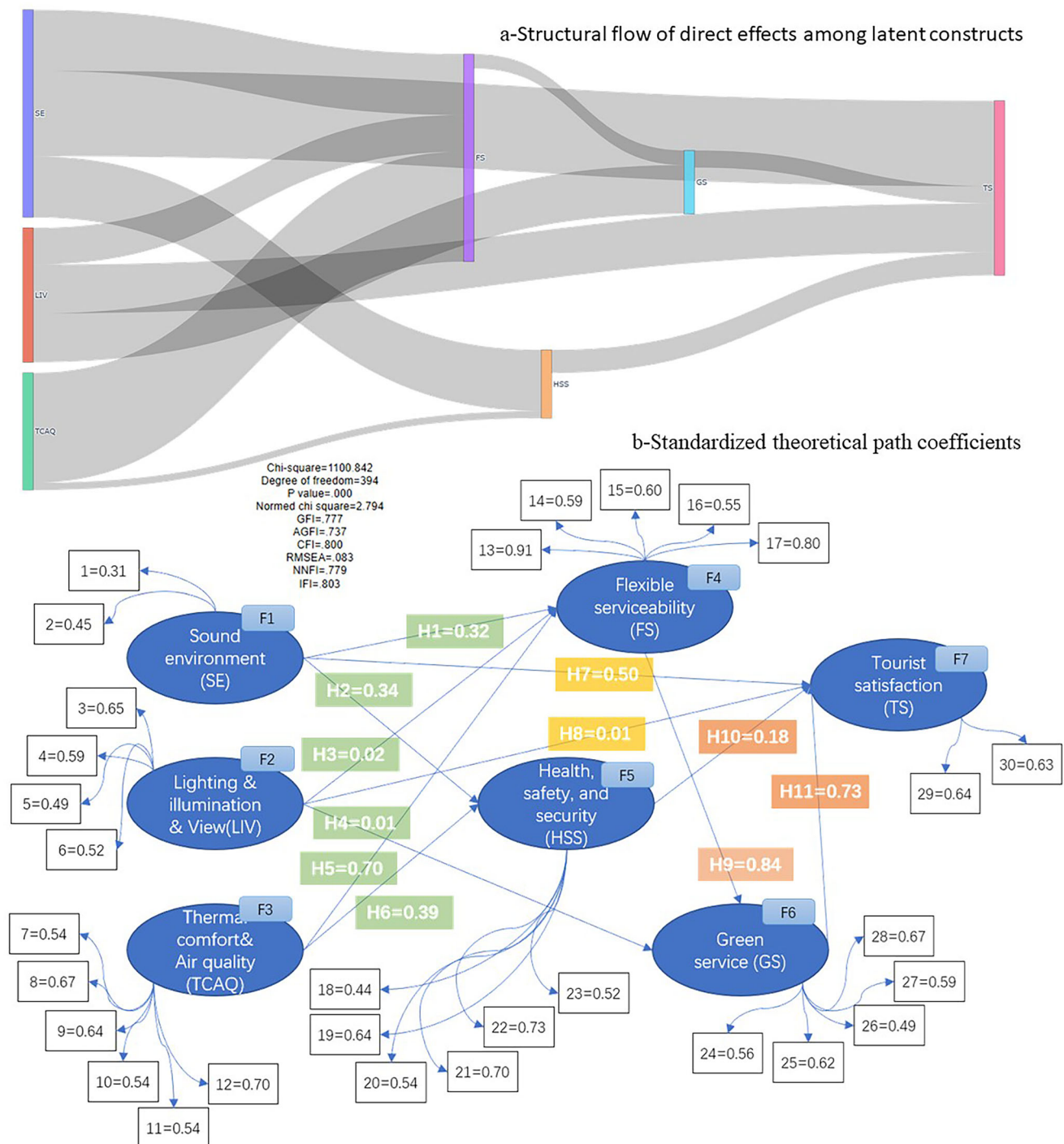


Fig. 3 Standardized theoretical path coefficients. **a** Structural flow diagram showing directed effects among latent constructs, with link widths proportional to standardized coefficients. **b** Standardized path-coefficient map highlighting dominant influences, including TCAQ $\rightarrow$ FS ($\beta = 0.90$) and SE $\rightarrow$ TS ($\beta = 0.70$), alongside moderate lighting effects ($\beta = 0.30$ – 0.40) and weaker safety and green-service paths ($\beta = 0.06$ – 0.19). Together, the panels summarize the hierarchical structure of direct effects driving tourist satisfaction.

sustainable hotel development, establishing a comprehensive model that links environmental factors with tourist satisfaction. From a theoretical perspective, the results of this study underscore the transformative potential of robotic 3D printing technology in the hotel industry. The analysis highlights how features such as sound insulation, lighting adaptability, air quality regulation, and safety measures significantly influence tourist satisfaction. These environmental qualities, supported by the hypotheses H7 and H8, emphasize the importance of

integrating advanced prefabrication technologies into hotel design to achieve superior guest experiences. Furthermore, the study demonstrates that service-related attributes, including flexible space layouts, green service quality, and robust health and safety measures, play a pivotal role in shaping guest satisfaction. These findings, supported by hypotheses H9, H10, and H11, reflect the interconnectedness of service quality and sustainability, providing a multidimensional ergonomic latent modeling.

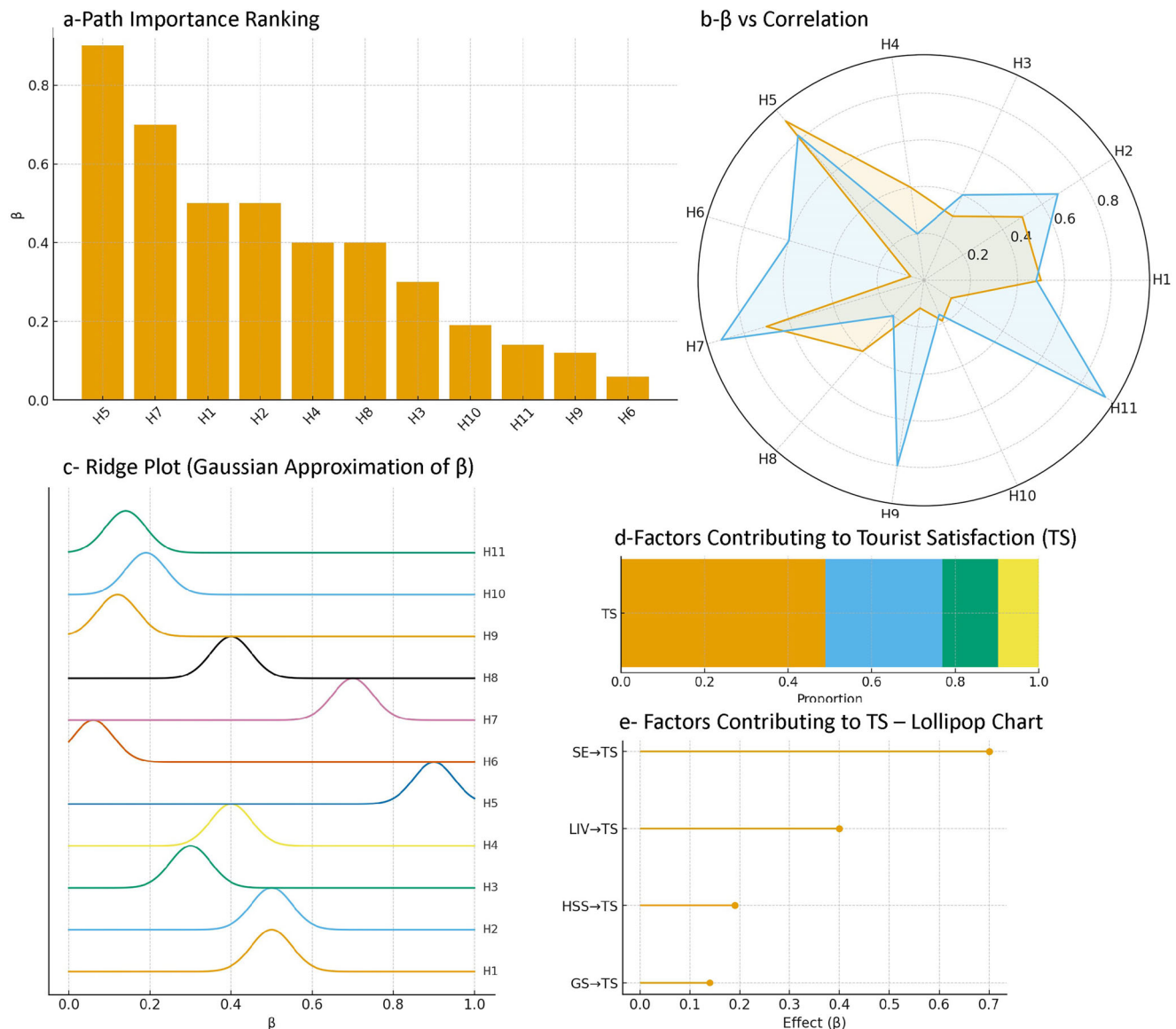


Fig. 4 Multidimensional evaluation of structural effects, correlational patterns, and construct contributions. **a** Ranking of standardized path coefficients (β) highlights TCAQ $\rightarrow$ FS ($\beta = 0.90$) and SE $\rightarrow$ TS ($\beta = 0.70$) as the dominant structural effects. **b** Dumbbell comparison of β and bivariate correlations (r) reveals strong convergence for H5 and H7, but marked inflation of r relative to β for H11 (0.92 vs 0.14) and H6 (0.60 vs 0.06), indicating mediation and shared variance. **c** Ridge density profiles show β values concentrated between 0.10 and 0.50 with a single extreme mode at 0.90. **d, e** Direct determinants of TS are led by SE (0.70), followed by LIV (0.40), HSS (0.19), and GS (0.14), as visualized via rose and lollipop representations.

From a practical standpoint, the implications of this research are multifaceted and relevant to various stakeholders. For hotel operators, adopting green prefabrication strategies informed by the study can lead to improved guest satisfaction and a stronger competitive position in the market. Features such as personalized modular designs, effective air quality management, and eco-friendly consumables offer practical avenues for achieving these outcomes. For policymakers, the study's findings provide a basis for refining industry standards and encouraging the adoption of sustainable construction technologies. Additionally, the technological insights highlight the importance of advancing robotic 3D printing techniques, particularly in material efficiency and modularity, which can further enhance the viability of green prefabrication methods. Furthermore, the application of robotic 3D printing prefabricated decoration not only supports sustainable practices in conventional hospitality settings but also opens new avenues for construction in extreme environments and even

extraterrestrial contexts, such as the Moon or Mars. By incorporating a human perception perspective, this research contributes valuable knowledge for the development of livable and resilient built environments in the most challenging and uncharted domains.

The results of this study highlight the synergistic potential of integrating ergonomic principles into robotic 3D printing-driven prefabricated building designs. By adopting a multidimensional latent modeling approach, the analysis captured both explicit performance indicators (e.g., material efficiency, energy savings) and latent variables related to user comfort and satisfaction. This methodological integration provides a more comprehensive understanding of how adaptive decoration strategies can be optimized to meet the dual objectives of environmental sustainability and human-centered design. Furthermore, the adaptability of prefabricated decorations offers a pathway for extending the applicability of these solutions to extreme environments—such as

high-humidity islands, polar regions, and even extraterrestrial habitats—where modularity, rapid deployment, and ergonomic considerations become critical for long-term habitability. These findings underscore the importance of combining advanced manufacturing technologies with human factors engineering to redefine sustainable hospitality and expand its frontiers beyond conventional contexts.

Despite its theoretical and empirical contributions, this study has several limitations that offer directions for future research. First, the use of cross-sectional survey data and structural equation modeling (SEM), while methodologically sound, may not fully capture the dynamic and complex nature of implementing robotic 3D printing in real-world hotel renovation projects. Longitudinal studies and experimental validation in operational contexts are recommended to strengthen causal inferences and assess long-term impacts. Second, the study's geographic concentration may limit the generalizability of its findings. Future research should conduct cross-regional and cross-cultural analyses to validate the proposed framework in different environmental, cultural, and regulatory contexts, thereby enhancing its global applicability. Third, the scalability and practical deployment of robotic 3D printing technologies in various hotel types and scales—especially in large hospitality chains—remain insufficiently explored. Investigating the technical, managerial, and infrastructural barriers to implementation will be crucial. Fourth, integrating emerging technologies such as digital twins and artificial intelligence (AI) into prefabrication workflows holds promise for enhancing design precision, energy efficiency, and user customization. Further research is needed to examine how these technologies can be synergistically combined to optimize operational and environmental outcomes. Lastly, future studies should assess the economic feasibility and cost-effectiveness of adopting robotic prefabrication, including lifecycle cost analysis and return on investment metrics. Additionally, exploring supportive policy mechanisms and incentive structures can facilitate industry-wide adoption and contribute to the broader advancement of low-carbon and aesthetic hotel renovations in alignment with global sustainability goals.

Conclusion

In conclusion, this study underscores the significant potential of green prefabrication technologies, particularly those leveraging robotic 3D printing, in advancing sustainable practices within the hospitality industry. The findings validate the proposed model, which demonstrates how enhancing environmental quality and service flexibility contributes to higher tourist satisfaction. These results align with global sustainability objectives and offer actionable insights for stakeholders seeking to integrate sustainability into hotel design and operations. Achieving widespread adoption of these technologies will require collaborative efforts among industry players, policymakers, and technology providers. Standardizing green prefabrication practices and fostering innovation in robotic 3D printing are critical steps toward realizing these goals. Future research should prioritize cross-cultural studies, economic evaluations, and technological advancements to ensure the scalability and practicality of green prefabrication strategies. This study not only provides a foundation for further exploration but also sets a benchmark for integrating cutting-edge technology with sustainability in the hospitality industry.

Data availability

Data will be made available from the corresponding author upon request.

Received: 22 January 2025; Accepted: 3 February 2026;

Published online: 04 March 2026

Notes

- 1 National Health Commission of the PRC et al. Notice on the Measures for the Ethical Review of Life Science and Medical Research Involving Humans (27 February 2023). <https://www.nhc.gov.cn/qjjys/c100016/202302/6b6e447b3edc4338856c9a652a85f44b.shtml>
- 2 Personal Information Protection Law of the PRC (effective 1 November 2021). https://www.cac.gov.cn/2021-08/20/c_1631050028355286.htm; GB/T 35273-2020 Information security technology—Personal information security specification. <https://openstd.samr.gov.cn/bzgk/gb/newGbInfo?hcno=4568F276E0F8346EB0FBA097AA0CE05E>

References

- 2025 China Hotel Industry Development Report: Seizing opportunities in cyclical patterns. (n.d.). Retrieved 28 July 2025, from <https://big5.cctv.com/gate/big5/culture-travel.cctv.com/2025/04/23/ARTI0nTtKzf6LtpX3gPuoPv250423.shtml>
- Abdollahi A, Arefi H, Malihi S, Maboudi M (2023) Progressive model-driven approach for 3D modeling of indoor spaces. *Sensors*, 23. <https://doi.org/10.3390/s23135934>
- Abdullah A, Dunn M, Yu K (2024) Robotic 3D printing of continuous fiber reinforced thermoset composites. *Adv Mater Technol*. <https://doi.org/10.1002/admt.202400839>
- Arenhart RS, Martins T, Ueda RM, Souza AM, Zanini RR (2024) Energy use and its contributors in hotel buildings: a systematic review and meta-analysis. *PloS One* 19(10):e0309745
- Azzaoui H, Dyba Y (2022) Modern trends in the design and construction of hotels in Morocco. *Archit Stud*. <https://doi.org/10.56318/as2022.01.007>
- Badraoui B, Smith S (2024). Innovative interiors: an evaluation of guest perceptions and booking responses to robotic furniture in hotel rooms. *J Hosp Tour Insights*. <https://doi.org/10.1108/jhti-05-2024-0479>
- Bao C, Moeinnia H, Kim T-H, Lee W, Kim WS (2023) 3D structural electronics via multi-directional Robot 3D printing. *Adv Mater Technol*. 8. <https://doi.org/10.1002/admt.202201349>
- Bedarf P, Szabo A, Zanini M, Dillenburger B (2023) Robotic 3D printing of geopolymer foam for lightweight and insulating building elements. *3D Print Addit Manuf* 11:1–9. <https://doi.org/10.1089/3dp.2023.0183>
- Bian J, Liu C, Zuo C, Hao J, Ma W, Duan B, Chen C, Liu J (2024) Reducing carbon emissions from prefabricated decoration: a case study of residential buildings in China. *Buildings* 14(2):550
- Cai G, Guo X, Sun Y (2025) Prefabricated building construction in materialization phase as catalysts for hotel low-carbon transitions via hybrid computational visualization algorithms. *Sci Rep* 15(1):7734. <https://doi.org/10.1038/s41598-025-92200-8>
- Cai G, Hong Y, Xu L, Gao W, Wang K, Chi X (2020) An evaluation of green ryokans through a tourism accommodation survey and customer-satisfaction-related CASBEE-IPA after COVID-19 pandemic. *Sustainability* 13(1):145
- Cai G, Liu D, Wu Z (2025) Multidimensional algorithms-based carbon efficiency model of building geometric 3D ratios for prefabricated 3D printing design and construction. *Npj Clean Energy* 1(1):5. <https://doi.org/10.1038/s44406-025-00005-5>
- Cai G, Lou Y, Lu F (2025) AI enhancing prefabricated aesthetics and low carbon coupled with 3D printing in chain hotel buildings from multidimensional neural networks. *Sci Rep* 15(1):13229. <https://doi.org/10.1038/s41598-025-97858-8>
- Cai G, Sun L, Liu D, Xu B, Mo Z (2025) Potential of indoor room 3D ratio in reducing carbon emissions by prefabricated decoration in chain hotel buildings via multidimensional algorithm models for robot in-situ 3D printing. *J Build Eng* 101: 111757. <https://doi.org/10.1016/j.jobte.2024.111757>
- Cai G, Wang J, Lue A, Xu S, Wu Q, Liu K, Gao T, Du P, Lei B (2023) From pollution to green and low-carbon island revitalization: implications of exhibition-driven Sustainable Tourism (Triennale) for SDG 8.9 in Setouchi. *Processes* 11(2):623
- Cai G, Xu B, Lu F, Lu Y (2023) The promotion strategies and dynamic evaluation model of exhibition-driven sustainable tourism based on previous/prospective tourist satisfaction after COVID-19. *Eval Program Plan* 101:102355
- Cai G, Xu L, Gao W (2021) The green B&B promotion strategies for tourist loyalty: Surveying the restart of Chinese national holiday travel after COVID-19. *Int J Hosp Manag* 94:102704
- Cai G, Zhang M, Zhang X, Xi H, Chen Z, Liu C, Liu K, Liu X, Xu S, Yu Z (2023) Promoting Green Buildings and Low-Carbon Design Strategies of Green B&B Rooms for Sustainable Tourism after COVID-19. *Land* 12(3):633

- Cai G, Zou B, Chi X, He X, Guo Y, Jiang W, Wu Q, Zhang Y, Zhou Y (2023) Neighborhood spatio-temporal impacts of SDG 8.9: The case of urban and rural exhibition-driven tourism by multiple methods. *Land* 12(2):368
- Çallı BA, Çallı L, Çallı DS, Çallı F (2022) The impact of different types of service robots usage in hotels on guests' intention to stay. *J Hosp Tour Technol*. <https://doi.org/10.1108/jhtt-09-2021-0266>
- Chan W (2012) Energy benchmarking in support of low carbon hotels: Developments, challenges, and approaches in China. *Int J Hosp Manag* 31(4):1130–1142
- Cheng C, Gao R, Xu G, Liu J, Ran K (2022) Research on prefabricate-decoration integration technology based on prefabricated bathroom. *IOP Conf Ser: Earth Environ Sci* 983. <https://doi.org/10.1088/1755-1315/983/1/012105>
- Dong H, Weng T, Zheng K, Sun H, Chen B (2024) Review: application of 3D Printing Technology in Soft Robots. *3D Print Addit Manuf* 11(3):954–976. <https://doi.org/10.1089/3dp.2023.0127>
- Geng Z, Le W, Guo B, Yin H (2023) Analysis of factors affecting visual comfort in hotel lobby. *PLOS ONE*, 18. <https://doi.org/10.1371/journal.pone.0280398>
- Guamán-Rivera R, Martínez-Rocamora A, García-Alvarado R, Muñoz-Sanguinetti C, González-Böhme L, Auat-Cheein F (2022) Recent developments and challenges of 3D-printed construction: a review of research fronts. *Buildings*. <https://doi.org/10.3390/buildings12020229>
- Gündüz G, Özkar M (2024) A process-based framework for adaptable modules in robotic clay 3D printing. *Int J Archit Comput* 22:45–61. <https://doi.org/10.1177/14780771231225698>
- Gupta R, Dantsiou D (2013) Understanding the gap between 'as designed' and 'as built' performance of a new low carbon housing development in UK. Sustainability in Energy and Buildings: Proceedings of the 4th International Conference in Sustainability in Energy and Buildings (SEB' 12), 567–580
- Ho CY (2022) Applications of Robot Staff in Hotels. <https://doi.org/10.31234/osf.io/xd8hm>
- Hong Y, Cai G, Mo Z, Gao W, Xu L, Jiang Y, Jiang J (2020) The impact of COVID-19 on tourist satisfaction with B&B in Zhejiang, China: An importance-performance analysis. *Int J Environ Res Public Health* 17(10):3747
- Jeong JY, Hyun SS (2019) Roles of passengers' engagement memory and two-way communication in the premium price and information cost perceptions of a luxury cruise. *Tour Manag Perspect* 32: 100559. <https://doi.org/10.1016/j.tmp.2019.100559>
- Jiayue SD, Rotimi J, Zuo K (2024) Supply Chain Landscape of 3D printed buildings: a stakeholder decision support framework. *Buildings*. <https://doi.org/10.3390/buildings14061811>
- Jing X, Lv D, Xie F, Zhang C, Chen S, Mou B (2022) A robotic 3D printing system for supporting-free manufacturing of complex model based on FDM technology. *Ind Robot* 50:314–325. <https://doi.org/10.1108/ir-05-2022-0136>
- Kaifosh P, Reardon TR, Allen BD, Anderson C, Arnoud S, Arora R, Atray M, Awad L, Ayerbe F, Baker C, Baker N, Barachant A, Bard P, Beltran WP, Berenzweig A, Bhasin R, Bienkowski J, Bittner S, Boegner L, ... Labs, CATR (2025) A generic non-invasive neuromotor interface for human-computer interaction. *Nature*. <https://doi.org/10.1038/s41586-025-09255-w>
- Kanthimathi T, Rathika N, Fathima ASR, Srinivasan S & R, T (2024) Robotic 3D Printing for Customized Industrial Components: IoT and AI-Enabled Innovation. 2024 14th International Conference on Cloud Computing, Data Science & Engineering (Confluence), 509–513. <https://doi.org/10.1109/Confluence60223.2024.10463472>
- Kim T, Kang G, Lee D, Shim D (2024) Development of an Indoor Delivery Mobile Robot for a Multi-Floor Environment. *IEEE Access* 12:45202–45215. <https://doi.org/10.1109/ACCESS.2024.3381489>
- Kyriazos T, Poga-Kyriazou M (2023) Applied psychometrics: Estimator considerations in commonly encountered conditions in CFA, SEM, and EFA practice. *Psychology* 14(5):799–828
- Lertputtak S, Treepob A, Samokhin D (2023) The improvement of service quality of service quality in prefabricated steel structure construction process. *Scholar: Hum Sci* 15(2):96–107
- Li T (2023) Building a robot path optimization in prefabricated buildings. 2023 IEEE 2nd International Conference on Electrical Engineering, Big Data and Algorithms (EEBDA), 1256–1261. <https://doi.org/10.1109/EEBDA56825.2023.10090581>
- Liu C, Yang Y, Zhao X, Xu X, Hao J, Wenting (2023) Reducing carbon emissions by using prefabricated decoration floor systems. *J Green Build*. <https://doi.org/10.3992/jgb.18.1.119>
- Lohakare H, Chainde T, Jadhav P, Rahangdale K, Wanjari D, Ikhar PA (2024) Design and Development of Robotic Arm Control by Human Hand. *Int J Res Publ Rev*. <https://doi.org/10.55248/gengpi.5.0324.07104>
- Mo Z, Gao T, Qu J, Cai G, Cao Z, Jiang W (2022) An empirical study of carbon emission calculation in the production and construction phase of a prefabricated office building from Zhejiang, China. *Buildings* 13(1):53
- Mo Z, Gao T, Qu J, Cai G, Cao Z, Jiang W (2023) An empirical study of carbon emission calculation in the production and construction phase of a prefabricated office building from Zhejiang, China. *Buildings* 13(1):53
- Mohandes SR, Abdelmageed S, Hem S, Yoo JS, Abhayajewea T, Zayed T (2022) Occupational Health and Safety in Modular Integrated Construction projects: The case of crane operations. *J Clean Prod* 342:130950
- Mostafaei F, Tahsildoust M, Zomorodian Z (2021) Energy efficiency and carbon emission in high-rise buildings: A review (2005–2020). *Build Environ* 206:108329
- Ordoñez-Ávila J, Lanza RE, Lázaro-Cárdenas SJ (2023) 3D Printed Structures for Under Water Robots Design. Proceedings of the 9th World Congress on Mechanical, Chemical, and Material Engineering. <https://doi.org/10.11159/icmie23.137>
- Ostapska K, Rütter P, Loli A, Gradeci K (2024) Design for Disassembly: A systematic scoping review and analysis of built structures Designed for Disassembly. *Sustain Prod Consum* 48:377–395
- Parisi, F, Sangiorgio, V, Parisi, N, Mangini, A, Fanti, M & Adam, J (2023). A new concept for large additive manufacturing in construction: tower crane-based 3D printing controlled by deep reinforcement learning. *Construction Innovation*. <https://doi.org/10.1108/ci-10-2022-0278>
- Peng J, Feng Y, Zhang Q, Liu X (2023) Multi-objective integrated optimization study of prefabricated building projects introducing sustainable levels. *Sci Rep* 13(1):2821
- Pollák M, Kočíško M, Török J (2024) Design of a robotic 3D printing system. *J Phys: Conf Ser*. <https://doi.org/10.1088/1742-6596/2805/1/012001>
- Prianto E, Herianto Herliansyah M, Pramono H, Husna A, Adam R, Raditya A, Lda S (2022) FDM Based Custom 3D printer development in robotic arm mechanical prototype printing. *J Phys: Conf Ser*, 2406. <https://doi.org/10.1088/1742-6596/2406/1/012005>
- Puzatova A, Shakor P, Laghi V, Dmitrieva M (2022) Large-scale 3D printing for construction application by means of robotic arm and Gantry 3D printer: a review. *Buildings*. <https://doi.org/10.3390/buildings12112023>
- Qiao J, Zhang X, Hao C, Liu S, Zhang Y, Xing K, Yang P (2023) Post-occupancy evaluation of the actual performance of a low-carbon building. *Energy Rep* 10:228–243
- Rafsanjani A (2022) Turning materials into robots. *Science* 376:1272–1273. <https://doi.org/10.1126/science.abq4102>
- Raphael B, Senthilnathan S, Patel A, Bhat S (2023) A review of concrete 3D printed structural members. 8. <https://doi.org/10.3389/fbuil.2022.1034020>
- Shan J, Cai G, Lin A, Xu B (2023) Improvement strategies for enhancing user satisfaction with spatial publicness in privately owned public open space in post COVID-19: a case from Italy. *Buildings* 13(7):1773
- Shi W, Cai G, Jiang L, Liu Y, Cui Y-T (2025) A Multidimensional Index for inclusive age-friendly transit design: development and practical scenarios. *Sage Open* 15(4):21582440251396640
- Sigahi TFAC, Thatcher A, Bolis I, Rampasso IS, Salatoro P, Anholon R (2024) Actionable insights for human factors and ergonomics to achieve the Sustainable Development Goals: mapping past advancements and proposing future strategic paths. *Ergonomics*, 1–22
- Tran LN, Cai G, Gao W (2023) Determinants and approaches of household energy consumption: A review. *Energy Rep* 10:1833–1850
- Tran LN, Gao W, Lam PM, Cai G (2025) A hybrid modeling approach to predicting HVAC demand in Japanese houses using physics-based simulation and Artificial Neural Networks. *Therm Sci Eng Prog* 67:104125. <https://doi.org/10.1016/j.tsep.2025.104125>
- Walczyk G, Ożadowicz A (2024) Building information modeling and digital twins for functional and technical design of smart buildings with distributed iot networks—review and new challenges discussion. *Future Internet* 16(7):225
- Wei J, Ge B, Zhong Y, Lee TL, Zhang Y (2024) Comparative analysis of embodied carbon in modular and conventional construction methods in Hong Kong. *Sci Rep* 14(1):23603
- Wong CTC, So ATP, Platten A (2015) Building low carbon high rise buildings in a sub-tropical climate: case study. In *Building Sustainable Futures: Design and the Built Environment* (pp. 157–182). Springer
- Xia M, Zhang Y (2022) Linear and nonlinear relationships: a hybrid SEM-neural network approach to verify the links of online experience with luxury hotel branding. *J Hosp Tour Insights* 5(5):1062–1079
- Xu J, Li M, Huang D, Wei Y, Zhong S (2022) A comparative study on the influence of different decoration styles on subjective evaluation of hotel indoor environment. *Buildings*. <https://doi.org/10.3390/buildings12111777>
- Xue W, Sun Z, Ye H, Liu Q, Jian B, Wang Y, Fang H, Ge Q (2024) Rigid-flexible coupled origami robots via multimaterial 3D printing. *Smart Mater Struct*, 33. <https://doi.org/10.1088/1361-665X/ad212c>
- Yang X (2024) Tourist Perceived Destination Innovativeness (TPDI) and its Impacts on Tourist Experience and Tourist Satisfaction-The Case Study of Hunan (China). University of Plymouth
- Ye X, Li-ke T, Qin-en G (2023) Low carbon design strategy of regional airport terminal based on building carbon emission calculation standard. *Int Civil Eng Archit Conf*, 641–653
- Zhang XH, Chang BG, Wu KS (2022) COVID-19 Shock, Financial flexibility, and hotels' performance nexus. *Front Public Health*, 10. <https://doi.org/10.3389/fpubh.2022.792946>

Živković M, Žujović M, Milošević J (2023) Intelligent tools for large-scale 3D printing: a scoping review. *J Faculty Civil Eng Archit.* <https://doi.org/10.62683/zrgaf38.88-99>

Author contributions

G.C. wrote the main manuscript text (original draft, review, and revision), project administration, methodology, and visualization. G.C. prepared figures and tables. B.X. and Z.H. contributed to formal analysis. Z.G., S.H., and F.L. contributed to the methodology and analysis. Y.X. and Z.R. contributed to conceptualization and analysis. B.J.D. and W.G. provided supervision. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

An exemption (waiver of ethics committee review) was granted by the School of Civil Engineering and Architecture, Zhejiang Sci-Tech University, on 15 January 2025 (Exemption/Reference No.: 20250115-1). The exemption was granted because the study was a non-interventional, anonymous online questionnaire on hotel interior decoration and tourists' perceptions, collected no personally identifiable information and no sensitive personal information, and posed no more than minimal risk to participants. All procedures were conducted in accordance with the ethical principles of the Declaration of Helsinki (or equivalent standards) and applicable regulations and data-protection requirements. In line with the scope defined in China's Measures for the Ethical Review of Life Science and Medical Research Involving Humans (issued 27 February 2023), which governs ethics review for life-science and medical research involving humans, the study was handled as exempt by the Approval Body.¹ Data handling complied with the Personal Information Protection Law (PIPL) and relevant national standards for personal information protection.²

Informed consent

Informed consent was obtained from all participants using an implied electronic consent procedure via the Wenjuanxing platform. During the survey period (2 December 2024 to 17 December 2024), an information sheet prepared by the research team was presented

on the first page of the questionnaire and described the study purpose, why the research was being conducted, voluntary participation, anonymity of responses, foreseeable risks (none beyond minimal risk), data-handling procedures, and the option to withdraw by exiting the survey at any time before submission. Consent was indicated by each adult participant (aged ≥ 18 years) voluntarily proceeding to complete and submit the anonymous questionnaire. Consent covered participation, use of anonymized responses for research analysis, and publication of aggregated results. The research team analyzed only anonymized responses and reported results only in an aggregated/anonymized form. We confirmed that informed consent was obtained from all subjects.

Additional information

Correspondence and requests for materials should be addressed to Gangwei Cai, Zhenwei Guo, Feidong Lu or Ziming Ren.

Reprints and permission information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2026